

Bit-Level Logic Instructions

Fundamentals of Ladder Diagram Programming

1 Ladder Diagram Language

Ladder diagram (LD) is a symbolic set of instructions used to create controller programs. These instructions represent commands that the PLC processor must execute to achieve desired control logic.

Symbolic Instructions

Each instruction in ladder logic is a command to perform a specific operation. They are arranged in "rungs" and entered into the PLC's memory. Every instruction relates to a **single bit** of memory, identified by a unique memory address.

2 Fundamental Symbols

To translate traditional relay control logic into symbolic PLC logic, three fundamental symbols are used:

1. **Examine If Closed (XIC)**: Corresponds to a normally open contact.
2. **Examine If Open (XIO)**: Corresponds to a normally closed contact.
3. **Output Energize (OTE)**: Corresponds to a relay coil or load.

3 Logic Instruction Status Table

The status of an instruction (TRUE or FALSE) depends directly on the value of its associated bit in the data table.

Data Table Bit Status	XIC (☐)	XIO (☐ / ☐)	OTE ((☐))
Logic 0 (OFF)	False	True	False
Logic 1 (ON)	True	False	True

Table 1: Relationship between Memory Bits and Instruction Status

4 Instruction Deep Dive

4.1 XIC — Examine If Closed (Logic Flag: 1)

Also known as the **Examine-on** instruction.

- **Process:** The processor examines the memory bit linked to an input device or internal flag.
- **Status:** If the bit is **1**, the instruction is **TRUE** (providing logical continuity).
- **Analogy:** Acts like a switch that is "closed" only when the input is active.

4.2 XIO — Examine If Open (Logic Flag: 0)

Also known as the **Examine-off** instruction.

- **Process:** Acts like a **normally closed** relay contact in software.
- **Status:** If the bit is **0**, the instruction is **TRUE**. If the bit is **1**, it becomes **FALSE**.
- **Analogy:** The processor "examines" to see if the contact remains open (i.e., bit is 0).

4.3 OTE — Output Energize

- **Condition:** If a continuous path of **TRUE** instructions exists in a rung (Logical Continuity), the OTE becomes **TRUE**.
- **Action:** When **TRUE**, it sets its memory bit to **1**; when **FALSE**, it sets it to **0**.

5 Combined Logic Diagram Example

Consider a machine circuit with a **Start Button** (I:1/0) and an **Emergency Stop** (I:1/1).

Rung Logic Configuration

Rung Layout: ---[XIC I:1/0]---[XIO I:1/1]---(OTE O:2/0)---

Logic Outcome Table:

Condition	Start (XIC)	E-Stop (XIO)	Output (OTE)	Reason
Normal Stop	0 (False)	0 (True)	0	Start not pressed.
Running	1 (True)	0 (True)	1	Start pressed, E-Stop OK.
Emergency	1 (True)	1 (False)	0	E-Stop breaks continuity.

Logical Continuity Summary

For an output to turn ON, there must be a continuous path of **TRUE** instructions from the left rail to the output. **XIC** provides continuity on a "1" bit; **XIO** provides continuity on a "0" bit.